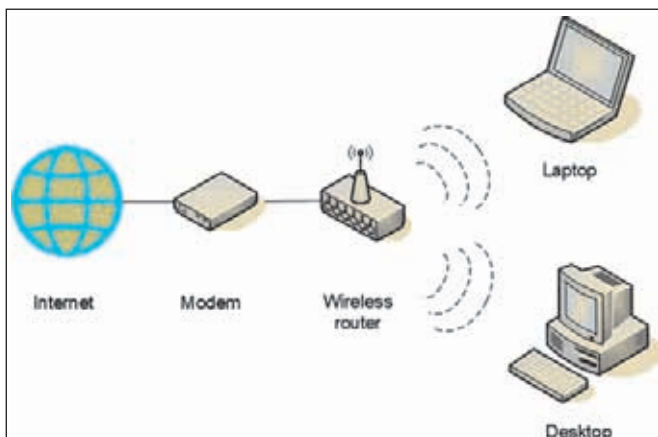




Wi-fi transmissions from one system to another

6.2 How Wi-Fi works

Transmission of Signals: Wi-Fi is a wireless network enabling technology that uses radio waves. We've covered RF transmissions and how to send and receive basic radio signals in the earlier chapters of this Fast Track and hence we shall avoid pursuing this further here. Wireless communication across Wi-Fi is a lot like the standard two-way radio communication.

First, the wireless adapter of a computer translates data into a radio signal that it transmits using an antenna. Then, a wireless router receives the signal and decodes it. The decoded information is sent to the internet using a physical, wired Ethernet connection. Of course, the reverse process is also duly carried out. The radios for Wi-Fi communication broadcast and receive at frequencies of 2.4 GHz or 5.0 GHz. The 2.4 GHz spectrum is the ISM spectrum (industrial, science and medical usage) and is an unlicensed part of the electromagnetic spectrum. Both 2.4 GHz and 5 GHz are much higher than the standard frequencies used for cell phones and other wireless devices. The higher frequency allows the signal to carry more data.

The way data is transmitted depends on the protocol and specifications of the device. Since all Wi-Fi devices use the same 802.11 protocols, they should be compatible and interoperable.